

Roshan Raj

Curriculum Vitae

Upto Oct 2025

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Research Interest

To pursue advanced research in **photon-based quantum computing and information**.

Educational Details

2025-Present	M.Sc. Physics (Laser & Photonics) , Universität Hamburg (UHH), Hamburg, Germany. [Focus: Quantum optics, and quantum computing & information.]
2025	PhD Physics (<i>enrolled briefly</i>), IISER, Mohali, India. <i>Transitioned to focused specialization in Laser & Photonics at Universität Hamburg .</i>
2019-24	Integrated M.Sc. Physics , SV National Institute of Technology (SVNIT), Surat, India. CGPA: 9.37/10
2019	B.Sc. (hon.) Physics (<i>enrolled briefly</i>), Patliputra University, Patna, India. <i>Transitioned to 5 years Integrated M.Sc. in Physics at SVNIT Surat .</i>
2016-18	Senior Secondary School, Loyola High School, CBSE, Patna, India, 77.2%
2010-16	Secondary School, St. Paul's High School, ICSE, Patna, India, 91.6% ,

Other Qualifications

2025	Joint UGC-CSIR National Eligibility Test (NET) Physical Science.
2023-25	Graduate Aptitude Test in Engineering (GATE) Physics.
2018-19	Joint Entrance Examination (JEE Main)
2016	National Talent Search Examination (NTSE).

Research Experiences

Master's Dissertation

- Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum
Supervisor: Dr. Vikash K Ojha, SVNIT-Surat; [Jul, 2023-May, 2024]–URL: [G Drive](#)
Brief: An attempt on the reconstruction of relativistic quantum mechanics, thermodynamics, etc.

Institute Mini-projects

- (3) New light on the concepts of Observable and indeterminacy in the quantum realm
Guide: Dr. Vikash K Ojha, SVNIT-Surat– UG Project III, [Jul-Dec, 2022]–URL: [G Drive](#)
Keywords: Quantum to Classical Transition
- (2) On the investigation of two non-neutral static bodies
Guide: Dr. Sharad K Yadav, SVNIT-Surat– UG Project II, [Jan-May, 2022]– DOI: [arXiv.2209.10641](#)
Keywords: Gravitation, Electromagnetism
- (1) Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle
Guide: Dr. Himanshu Pandey, SVNIT-Surat– UG Project I, [Jul-Dec, 2021]–DOI: [10.31219/osf.io/ukx9t](#)
Keywords: Gravitation, Electromagnetism

Independent Works

- (6) On one of the applications of Electrodynamics and Crystals. 2025. URL: [G Drive](#).
Brief: This work is based on self inquiring sequentially through queries and trying to resolve them to build a mind map on topics of optics and light-matter interaction.
- (5) On Einstein's Theory of Gravitation. 2025. URL: [G Drive](#).
Brief: This work is based on the lecture notes on general relativity and cosmology.
- (4) The Circle: 7 different perspectives. Magazine (AMaThing 5.0). 2024. URL: [G Drive](#)
Brief: An exploration of the mathematical concept of a circle through various lenses of mathematical disciplines (geometry, algebraic topology, complex numbers, knot theory, etc.)
- (3) Obtaining the Klein-Gordon Wave Equation Without Using the Quantum Operators. OSF Preprints. 2023. DOI: [10.31219/osf.io/6hmyf](#).
Brief: We claim that Klein-Gordon (KG) equation should be termed as "Wave equation" and Schrödinger equation (also, Dirac's equation) is proposed to be called as "Governing law". The basis of categorisation is discussed.
- (2) Reviewing Observables in Classical and Quantum Mechanics. OSF Preprints. 2023. DOI: [10.31219/osf.io/p2ufx](#).
Brief: It presents the evolution of the quantum observables and distinctly analyses the meaning of observable in context to both the classical and the quantum physics, emphasising 1925's W. Heisenberg realization.
- (1) An Open-ended Story on Quantization. OSF Preprints. 2023. DOI: [10.31219/osf.io/92bvp](#).
Brief: A write-up impressing on the evolutionary track of quantization of physical quantities such as Energy, momentum, and angular momentum.

Internships

- (3) प्रमात्रिक गतिकी के चित्र [*The pictures of quantum dynamics* (Hindi)]
Guide: Dr. Vikash K Ojha, SVNIT, Surat, [May-Jul, 2022]—URL: [G Drive](#)
Brief: Physics Education for Hindi speaking readers. The work explores 3 quantum dynamics methods and explains analytical mechanics, quantum mechanics, and ontological view. It aims to aware the common public about basics of quantum dynamics and its historical context.
- (2) The front-end development of a dummy e-commerce website
Guide: Dr. Santosh Prabhakar, Internship Studio, Pune, [Jun-Jul, 2020]
Brief: Learning included using HTML5, CSS3 frameworks and training for the web.
- (1) Project, Building the student information management system using C language
Guide: Dr. Balu Parne, SVNIT, Surat, [Feb-May, 2020]
Brief: Learning included using C languages and its application in file handling.

Responsibilities & Positions

08/22-03/23	Student Coordinator: QUANTA Seminar Series, Department of Physics, SVNIT Surat
11/22-01/23:	Design-Infra Co-Head & Org. Member: <i>Manoj Memorial Night Cricket Tournament</i> (MM-NCT). SVNIT Surat.
10/21-07/22:	Joint Academic Affairs Secretary: <i>Academic Affairs Council</i> . SVNIT, Surat
06/20-07/22	Co-convener (03/21) & Member. <i>Society for Cultivation of Science and Humanities</i> (SCOSH), Student Chapter, SVNIT Surat
09/19-07/20	Student Volunteer: <i>Unnat Bharat Abhiyan</i> (UBA), SVNIT Surat

Workshops, Seminars & Talk

11/21:	Given a talk on <i>Unification in Mathematics</i> , IntERAct Seminar series, SVNIT, Surat
07/21:	Participated in <i>Quantum Fields, Geometry & Representation Theory</i> , ICTS-TIFR, Bangalore.
12/20:	Participated in National workshop on <i>Data analysis using MS Excel</i> , BBD NITM, Lucknow.
08/20:	Participated in National workshop on <i>MATLAB Tools And Applications</i> , BBD NITM, Lucknow.

Achievements

2016-25	NTS Scholarship of merit, Government of India, NCERT, New Delhi.
2020	Winner (Physics), InQuest 4.0 Researchathon, SCOSH [Student Chapter], SVNIT

Online Courses

08-09/24:	PFW Bangladesh: <i>School On Quantum Field Theory</i> , ICTP Italy, Et al.
05-06/24:	Algebraic Topology in Physics, Physics Latam, Prof. Pavel Putrov (ICTP).
08-09/21:	A Short Course in Math History, Prof. A. Viramani (CMI)
2019-2020:	Multiple online courses under Prof. H. C. Verma (IIT Kanpur): Electromagnetism, Special Relativity, Experimental Techniques, Quantum Mechanics.

Skills

Programme:	C, Python, MATLAB, Octave, MySQL, Mathematica
Softwares:	L ^A T _E X, MS Office, and use of GitHub
Web Design:	Basics of HTML5, CSS3
Other:	Canva, Notion
Languages:	English (fluent), Hindi (Native)

Other

I have always been an enthusiastic, independent seeker of truth. I wanted to explore mathematical beauty, ancient history, and philosophical ideas and stand up for social justice. My hobbies include composing poems (self-published two collections as **मनोविघ्न** (MaNoVighn) and **कालसंघर्ष** (KālSanGharsha)) and writing articles on online platform (Pratilipi).

Declaration: *I hereby declare that the information given above is true to the best of my knowledge.*

Digitally Signed: **Roshan Raj**

Date: 31/10/2025.